

Turunan Trigonometri

Pertemuan Ke-7

TURUNAN

Aturan Turunan
- F. trigono
- F. aljabar.

1. Aturan konstanta. Jika $f(x) = k$, maka $f'(x) = 0$
2. Aturan penjumlahan. Jika ada $f(x)$ dan $g(x)$, maka turunannya $f'(x) + g'(x)$
3. Aturan pangkat. Jika $f(x) = x^n$, maka $f'(x) = nx^{n-1}$
4. Aturan perkalian. Jika ada perkalian dua fungsi maka $(uv)' = u'v + uv'$
5. Aturan pembagian. Jika ada pembagian dua fungsi maka $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

TURUNAN FUNGSI TRIGONOMETRI

Rumus Turunan Trigonometri

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\csc x$	$-\csc x \cdot \cot x$
$\sec x$	$\tan x \cdot \sec x$
$\cot x$	$-\csc^2 x$
$\sin(ax+b)$	$a\cos(ax+b)$
$\cos(ax+b)$	$-a\sin(ax+b)$

$f(x) = \sin(ax+b)$

$f'(x) = a\cdot\cos(ax+b)$

$f(x) = \sin(2x+3)$; $ax+b \rightarrow a=2,\ b=3$

$f'(x) = 2\cdot\cos(2x+3)$ //

$f(x) = \cos(3x+1)$; $ax+b \rightarrow a=3,\ b=1$

$f'(x) = -3\sin(3x+1)$ //

$f(x) = \sin(ax-b)$

$f'(x) = a\cos(ax-b)$ //

$f(x) = \cos(ax-b)$

$f'(x) = -a\sin(ax-b)$

$f(x) = \cos(-2x-1)$; $ax-b \rightarrow a=-2,\ b=1,\ -a=2$

$f'(x) = 2\sin(-2x-1)$ //

$ax \oplus b \rightarrow a=-2,\ b=-1$

$-2x-1$

$f(x) = 5\sin x$

$f'(x) = 5\cos x$

$f(x) = -2\cos x$

$f'(x) = -2(-\sin x)$

$= 2\sin x$ //

$\forall x \rightarrow \sin x$

$\tan x = \frac{\sin x}{\cos x}$

$\sec x = \frac{1}{\cos x}$

$\sec^2 x = \frac{1}{\cos^2 x}$ //

TRIGONOMETRY LAWS AND IDENTITIES

TANGENT IDENTITIES

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\cot\theta = \frac{\cos\theta}{\sin\theta}$$

RECIPROCAL IDENTITIES

$$\csc\theta = \frac{1}{\sin\theta} \qquad \sin\theta = \frac{1}{\csc\theta}$$

$$\sec\theta = \frac{1}{\cos\theta} \qquad \cos\theta = \frac{1}{\sec\theta}$$

$$\cot\theta = \frac{1}{\tan\theta} \qquad \tan\theta = \frac{1}{\cot\theta}$$

PYTHAGOREAN IDENTITIES

$$\sin^2\theta + \cos^2\theta = 1$$

$$\tan^2\theta + 1 = \sec^2\theta$$

$$\cot^2\theta + 1 = \csc^2\theta$$

$\sin^2 x + \cos^2 x = 1$

$\cos^2 x + \sin^2 x = 1$

PERIODIC IDENTITIES

$$\sin(\theta + 2\pi n) = \sin\theta$$

$$\cos(\theta + 2\pi n) = \cos\theta$$

$$\tan(\theta + \pi n) = \tan\theta$$

$$\csc(\theta + 2\pi n) = \csc\theta$$

$$\sec(\theta + 2\pi n) = \sec\theta$$

$$\cot(\theta + \pi n) = \cot\theta$$

EVEN/ODD IDENTITIES

$$\sin(-\theta) = -\sin\theta$$

$$\cos(-\theta) = \cos\theta$$

$$\tan(-\theta) = -\tan\theta$$

$$\csc(-\theta) = -\csc\theta$$

$$\sec(-\theta) = \sec\theta$$

$$\cot(-\theta) = -\cot\theta$$

DOUBLE ANGLE IDENTITIES

$$\sin(2\theta) = 2\sin\theta\cos\theta$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$

$$= 2\cos^2\theta - 1$$

$$= 1 - 2\sin^2\theta$$

$$\tan(2\theta) = \frac{2\tan\theta}{1 - \tan^2\theta}$$

HALF ANGLE IDENTITIES

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$$

LAW OF COSINES

$$a^2 = b^2 + c^2 - 2bc\cos\alpha$$

$$b^2 = a^2 + c^2 - 2ac\cos\beta$$

$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$

PRODUCT TO SUM IDENTITIES

$$\sin\alpha\sin\beta = \frac{1}{2}[\cos(\alpha-\beta) - \cos(\alpha+\beta)]$$

$$\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$$

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]$$

$$\cos\alpha\sin\beta = \frac{1}{2}[\sin(\alpha+\beta) - \sin(\alpha-\beta)]$$

SUM TO PRODUCT IDENTITIES

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\sin\alpha - \sin\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$\cos\alpha - \cos\beta = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

LAW OF SINES

$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$$

LAW OF TANGENTS

$$\frac{a-b}{a+b} = \frac{\tan\left[\frac{1}{2}(\alpha-\beta)\right]}{\tan\left[\frac{1}{2}(\alpha+\beta)\right]}$$

$$\frac{b-c}{b+c} = \frac{\tan\left[\frac{1}{2}(\beta-\gamma)\right]}{\tan\left[\frac{1}{2}(\beta+\gamma)\right]}$$

$$\frac{a-c}{a+c} = \frac{\tan\left[\frac{1}{2}(\alpha-\gamma)\right]}{\tan\left[\frac{1}{2}(\alpha+\gamma)\right]}$$

SUM/DIFFERENCES IDENTITIES

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$

$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

MOLLWEIDE'S FORMULA

$$\frac{a+b}{c} = \frac{\cos\left[\frac{1}{2}(\alpha-\beta)\right]}{\sin\left(\frac{1}{2}\gamma\right)}$$

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COFUNCTION IDENTITIES

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$$

$$\csc\left(\frac{\pi}{2} - \theta\right) = \sec\theta$$

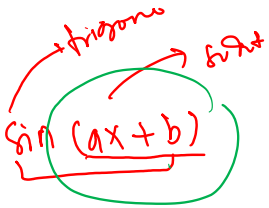
$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot\theta$$

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = \csc\theta$$

$$\cot\left(\frac{\pi}{2} - \theta\right) = \tan\theta$$

α	I					II				III				IV			
	0°	30°	45°	60°	90°	120°	135°	150°	180°	210°	225°	240°	270°	300°	315°	330°	360°
	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
$\sin\alpha$	0	$\frac{1}{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	1	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}\sqrt{3}$	-1	$-\frac{1}{2}\sqrt{3}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}$	0
$\cos\alpha$	1	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}\sqrt{3}$	-1	$-\frac{1}{2}\sqrt{3}$	$-\frac{1}{2}\sqrt{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	1
$\tan\alpha$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	td	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	td	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	0
$\csc\alpha$	td	2	$\sqrt{2}$	$\frac{2}{3}\sqrt{3}$	1	$\frac{2}{3}\sqrt{3}$	$\sqrt{2}$	2	td	-2	$-\sqrt{2}$	$-\frac{2}{3}\sqrt{3}$	-1	$-\frac{2}{3}\sqrt{3}$	$-\sqrt{2}$	-2	td
$\sec\alpha$	1	$\frac{2}{3}\sqrt{3}$	$\sqrt{2}$	2	td	-2	$-\sqrt{2}$	$-\frac{2}{3}\sqrt{3}$	-1	$-\frac{2}{3}\sqrt{3}$	$-\sqrt{2}$	-2	td	2	$\sqrt{2}$	$\frac{2}{3}\sqrt{3}$	-1
$\cot\alpha$	td	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	td	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	1	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	td



fungsi	f'(x)
sin x	→ cos x ✓
cos x	→ −sin x
tan x	→ sec²x = (sec x)²

LATIHAN SOAL

Carilah turunan dari fungsi berikut:

1. $y = \sin 5x + \cos 6x - \sin 3x$ → $y' = 5\cdot\cos 5x + 6\cdot(-\sin 6x) - 3(\cos 3x)$
   $y' = 5\cos 5x - 6\sin 6x - 3\cos 3x$ //

$y = \cos^2 x = (\cos x)^2$
$y' = (-\sin x)\cdot 2(\cos x)^1$
$= -2\sin x \cdot \cos x$ //

2. $y = x^2 \sin x$ (aturan pertama) f_1 f_2

(2) $y = x^2 \sin x$
$u = x^2$ $v = \sin x$
$u' = 2x$ $v' = \cos x$
$(uv)' = u'v + uv'$
$= (2x)(\sin x) + (x^2)(\cos x)$
$y' = 2x\cdot\sin x + x^2\cos x$ //

3. $f(x) = x \tan x$ dan tentukan $f'\left(\frac{\pi}{4}\right)$ ($f'(x) =$)

(3) $f(x) = x\tan x$
$u = x$ $v = \tan x$
$u' = 1$ $v' = \sec^2 x$
$(uv)' = u'v + uv'$
$= (1)(\tan x) + (x)(\sec^2 x)$
$f'(x) = \tan x + x\cdot\sec^2 x$
$f'\left(\frac{\pi}{4}\right) = \tan\left(\frac{\pi}{4}\right) + \left(\frac{\pi}{4}\right)\sec^2\left(\frac{\pi}{4}\right) = 1 + \left(\frac{\pi}{4}\right)(\sqrt{2})^2 = 1 + \frac{\pi}{2}$ //
($\sqrt{2}\sqrt{2} = 2$)

$y = \sin 5x$
$y' = 5\cos 5x$

$y = 5\cdot\sin x$
$y' = 5\cdot\cos x$

atau pakai → $y = \sin^5 x = (\sin x)^5$
$y' = 5(\sin x)^{5-1}\cdot\cos x$
$= 5\sin^4 x\cdot\cos x$ //

LATIHAN SOAL

Carilah turunan dari fungsi berikut:

4. $f(x) = 2\sin 3x \cdot \tan 2x$ dan tentukan $f'\left(\frac{\pi}{6}\right)$

5. $y = \tan x$

6. $f(x) = \frac{\sin 2x}{\cos x}$ dan tentukan $f'\left(\frac{\pi}{3}\right)$

④ $f(x) = 2\sin 3x \tan 2x$

$u = 2\sin 3x \quad v = \tan 2x$

$u' = 2 \cdot 3 \cdot \cos 3x \quad v' = 2 \cdot \sec^2 2x$

$u' = 6\cos 3x$

$(uv)' = u'v + uv'$

$= (6\cos 3x)(\tan 2x) + (2\sin 3x)(2\sec^2 2x)$

$f'(x) = 6\cos 3x \tan 2x + 4\sin 3x \sec^2 2x$

$f'\left(\frac{\pi}{6}\right) = 6\cos 3\left(\frac{\pi}{6}\right) \cdot \tan 2\left(\frac{\pi}{6}\right) + 4\sin 3\left(\frac{\pi}{6}\right) \cdot \sec^2 2\left(\frac{\pi}{6}\right)$

$= 6\cos\frac{\pi}{2} \cdot \tan\frac{\pi}{3} + 4\sin\frac{\pi}{2} \cdot \sec^2\frac{\pi}{3}$

$= (6)(0)(\sqrt{3}) + 4(1)(2)^2$

$= 0 + 16$

$f'\left(\frac{\pi}{6}\right) = 16$ //

⑤ $y = \tan x = \frac{\sin x}{\cos x}$

$u = \sin x \quad v = \cos x$

$u' = \cos x \quad v' = -\sin x$

$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

$= \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{(\cos x)^2}$

$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$

$= \frac{1}{\cos^2 x}$

$= \sec^2 x$ // (∴Terbukti) //

⑥ $u = \sin 2x \quad v = \cos x$

$u' = 2\cos 2x \quad v' = -\sin x$

$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$

$= \frac{(2\cos 2x)(\cos x) - (\sin 2x)(-\sin x)}{(\cos x)^2}$

$f'(x) = \frac{2\cos 2x \cdot \cos x + \sin 2x \cdot \sin x}{\cos^2 x}$

$= \frac{2\cos 2\left(\frac{\pi}{3}\right) \cdot \cos\left(\frac{\pi}{3}\right) + \sin 2\left(\frac{\pi}{3}\right) \cdot \sin\left(\frac{\pi}{3}\right)}{\cos^2\left(\frac{\pi}{3}\right)}$

$= 2 \cdot \left(-\frac{1}{2}\right) \cdot$

$=$

Kisi-kisi Soal UTS Reguler Pagi

1. Pertidaksamaan Nilai Mutlak
2. Domain dan Range Fungsi
3. Soal Cerita Limit Aljabar
4. Limit Trigonometri
5. Turunan Aljabar Fungsi Kuadrat
6. Turunan Trigonometri

LATIHAN SOAL
Pertidaksamaan Nilai Mutlak (Pert.3)

a. $|2x - 1| < 3$

b. $|3x - 5| \geq 1$

c. $|2x + 3| \geq |4x + 5|$

d. $|2x - 1| < 3$

e. $|2x + 1| > 5$

f. $|2x + 1| \leq |x - 3|$

g. $\left|\frac{x-1}{x+3}\right| \geq \frac{2}{3}$

h. $\left|\frac{x-4}{x+1}\right| < \frac{1}{2}$

LATIHAN SOAL
Domain dan Range Fungsi (Pert.4)

Tentukan daerah asal serta daerah hasil dari:

1. $f(x) = \sqrt{6 - 2x}$
2. $f(x) = 2x + 5$
3. $f(x) = 4 - 3x$
4. $f(x) = x^2 - 5x - 10$
5. $f(x) = -x^2 + 4x$
6. $f(x) = \frac{3}{x - 1}$
7. $f(x) = \frac{x + 2}{x - 4}$

Diketahui $f(x) = \sqrt{x}$ dan $g(x) = x - 2$.
Tentukan:

a. D_{fog} dan R_{fog}

b. D_{gof} dan R_{gof}

LATIHAN SOAL – LIMIT ALJABAR (Pert.5)

Soal 1

Kapasitas penyimpanan yang terpakai S (dalam GB) pada server cloud sebuah Perusahaan setelah x bulan dinyatakan dengan fungsi: $S(x) = \frac{x^2+2x-24}{x-4}$. Kapasitas penyimpanan yang terpakai pada bulan ke-4 ditentukan menggunakan limit. Hitunglah $\lim_{x\to 4} S(x)$ dan Interpretasikan hasil limit tersebut!

Soal 2

Kapasitas penyimpanan yang terpakai S (dalam GB) pada server cloud sebuah Perusahaan setelah x bulan dinyatakan dengan fungsi: $S(x) = \frac{x^2-x-20}{x-5}$. Kapasitas penyimpanan yang terpakai pada bulan ke-5 ditentukan menggunakan limit. Hitunglah $\lim_{x\to 5} S(x)$ dan Interpretasikan hasil limit tersebut!

Soal 3

Sebuah benda bergerak sepanjang garis lurus dengan posisi terhadap waktu t (dalam detik) dan S dalam meter dengan fungsi: $S(t) = \frac{t^2+4t-12}{t-2}$. Posisi benda pada saat t = 2 detik ditentukan menggunakan limit fungsi. Hitunglah $\lim_{t\to 2} s(t)$ dan Interpretasikan hasil limit tersebut!

LATIHAN SOAL – Limit Trigonometri(Pert.5 & 6)

Carilah nilai dari limit berikut:

1. $\lim_{x \to 0} \frac{2 \sin 4x}{7 \tan 7x}$
2. $\lim_{x \to \frac{\pi}{2}} (\sin x + \cos x)$
3. $\lim_{x \to 0} \frac{2x^2 + x}{\sin x}$
4. $\lim_{x \to 0} \frac{\tan 2x \cdot \tan 3x}{3x^2}$
5. $\lim_{x \to 0} \frac{1 - \cos x}{\sin x}$
6. $\lim_{x \to 0} \frac{\cos 2x}{\sin x - \cos x}$
7. $\lim_{x \to \frac{\pi}{6}} \frac{1}{2} \cos 12x$
8. $\lim_{x \to 0} \frac{2x \sin 3x}{\tan^2 5x}$
9. $\lim_{x \to 0} \frac{1 - \cos 2x}{x^2}$

LATIHAN SOAL

Turunan Aljabar (Pert.6)

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Carilah turunan berdasarkan definisi dari fungsi berikut:

1. $f(x) = 13x - 6$
2. $f(x) = x^2 - 2x$

Carilah turunan dari fungsi berikut:

1. $f(x) = x^3 + 3x^2 + 4$
2. $f(x) = (3x^2 - 5)(2x^4 - x)$
3. $f(x) = \frac{x+3}{x^2+1}$
4. $f(x) = \frac{5}{\sqrt{x}}$
5. $f(x) = \frac{3}{x^3} + x^{-4}$

LATIHAN SOAL
Turunan Trigonometri (Pert.7)

Carilah turunan dari fungsi berikut:

1. $y = \sin 5x + \cos 6x - \sin 3x$
2. $y = x^2 \sin x$
3. $f(x) = x \tan x$ dan tentukan $f'\left(\frac{\pi}{4}\right)$
4. $f(x) = 2 \sin 3x \,.\, \tan 2x$ dan tentukan $f'\left(\frac{\pi}{6}\right)$
5. $y = \tan x$
6. $f(x) = \frac{\sin 2x}{\cos x}$ dan tentukan $f'\left(\frac{\pi}{3}\right)$

Thank you

